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Inventor(s)

YAMAMOTO; Takashi et al.

WORK MACHINE AND OPERATION SYSTEM FOR WORK MACHINE

Abstract

A work machine includes a lower traveling body; an upper slewing body slewably mounted on the lower traveling body; an external microphone configured to collect sound around a work machine; an internal speaker configured to output sound to an operator of the work machine; an information transmission device configured to enable a worker around the work machine to recognize whether or not a voice of the worker has reached the operator of the work machine; and a processor, and a memory storing instructions that cause the processor to execute a process. The process includes collecting, by the external microphone, sound around the work machine; outputting, by the internal speaker, the sound to the operator of the work machine; and enabling, by the information transmission device, the worker around the work machine to recognize whether or not a voice of the worker has reached the operator of the work machine.

Inventors: YAMAMOTO; Takashi (Chiba, JP), MATSUHASHI; Yuiki (Chiba, JP), SHINKAI; Kazuma (Chiba, JP)

Applicant: SUMITOMO CONSTRUCTION MACHINERY CO., LTD. (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims priority to Japanese Patent Application No. 2024-037458, filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a work machine and an operation system for the work machine.

2. Description of Related Art

[0003] In the related art technologies, an excavator including a microphone for collecting a voice of a worker around the excavator is known.

SUMMARY

[0004] According to an embodiment of the present disclosure, a work machine is provided. The work machine includes: [0005] a lower traveling body; [0006] an upper slewing body slewably mounted on the lower traveling body; [0007] an external microphone configured to collect sound around a work machine; [0008] an internal speaker configured to output sound to an operator of the work machine; [0009] an information transmission device configured to enable a worker around the work machine to recognize whether or not a voice of the worker has reached the operator of the work machine; and [0010] a processor, and a memory storing instructions that cause the processor to execute a process, wherein the process includes [0011] collecting, by the external microphone, sound around the work machine; [0012] outputting, by the internal speaker, the sound to the operator of the work machine; and [0013] enabling, by the information transmission device, the worker around the work machine to recognize whether or not a voice of the worker has reached the operator of the work machine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a side view of a work machine according to an embodiment of the present disclosure.

[0015] FIG. 2 is a top view of the work machine illustrated in FIG. 1.

[0016] FIG. 3 is a view illustrating examples of configurations of an external sound collection device and an information transmission device attached to the work machine illustrated in FIG. 1.

[0017] FIG. 4 is a diagram schematically illustrating a configuration example of the work machine illustrated in FIG. 1.

[0018] FIG. 5 is a top view illustrating an inside of a cabin of the work machine illustrated in FIG. 1.

[0019] FIG. 6 is a perspective view illustrating the work machine boarded by an operator and a worker around the work machine.

[0020] FIG. 7 is a flowchart illustrating an example of a sequence of an information transmission process.

[0021] FIG. **8** is a flowchart illustrating an example of a proceed of a sound transmission process.
[0022] FIG. **9** is a flowchart illustrating an example of a proceed of a conversation notification process.
[0023] FIG. **10** is a top view of a work machine illustrating another configuration example of a work machine according to the embodiment of the present disclosure.
[0024] FIG. **11** is a schematic view illustrating a configuration example of an operation system according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0025] However, in the above-described related art technologies, the worker around the excavator serving as a work machine may not be able to determine whether or not his/her voice has reached the operator of the excavator.

[0026] Therefore, it is desirable for the worker to recognize that the voice of the worker has reached the operator of the work machine such as the excavator.

[0027] The work machine described above enables the worker to recognize that the voice of the worker around the work machine has reached the operator of the work machine.

[0028] Hereinafter, embodiments of the present disclosure will be described with reference to the accompanying drawings. The embodiments described below are merely examples and do not limit the invention. All features and combinations thereof in the embodiments of the present disclosure are not necessarily essential to the invention. In the drawings, the same or corresponding components are denoted by the same or corresponding reference numerals, and redundant description thereof may be omitted.

[0029] The work machine **100** according to the embodiment of the present disclosure is an excavator. The work machine **100** may be a machine other than the excavator, such as a crane, an asphalt finisher, or a forklift. In the illustrated example, the excavator as the work machine **100** is a digging machine including the bucket **6** as an end attachment, but may be an application machine such as a forestry machine including an end attachment other than the bucket **6**.

[0030] First, an overview of the work machine **100** will be described with reference to FIGS. **1** and **2**. FIG. **1** is a side view of the work machine **100**, and FIG. **2** is a top view of the work machine **100**.

[0031] In FIG. **1**, +X represents one direction of the X-axis constituting the three dimensional orthogonal coordinate system, and -X represents the other direction of the X-axis. In FIG. **2**, +Y represents one direction of the Y axis constituting the three dimensional orthogonal coordinate system, and -Y represents the other direction of the Y axis. In FIG. **1**, +Z represents one direction of the Z-axis constituting the three dimensional orthogonal coordinate system, and -Z represents the other direction of the Z-axis. In FIG. **1**, the +X side of the work machine **100** corresponds to the front side of the work machine **100**, and the -X side of the work machine **100** corresponds to the rear side of the work machine **100**. The +Y side of the work machine **100** corresponds to the left side of the work machine **100**, and the -Y side of the work machine **100** corresponds to the right side of the work machine **100**. The +Z side of the work machine **100** corresponds to the upper side of the work machine **100**, and the -Z side of the work machine **100** corresponds to the lower side of the work machine **100**. The same applies to the other drawings.

[0032] The work machine **100** includes a lower traveling body **1**, an upper slewing body **3** that is slewably mounted on the lower traveling body **1** via a slewing mechanism **2**, an attachment AT for performing various kinds of work, and a cabin **10**. The cabin **10** is also called an operation cabin or a cab. The front side of the work machine **100** (upper slewing body **3**) corresponds to a side on which the attachment AT is attached to the upper slewing body **3** when the work machine **100** is viewed from directly above along the slewing axis of the upper slewing body **3**. The left side, the right side, and the rear side of the work machine **100** (upper slewing body **3**) correspond to the left side, the right side, and the rear side, respectively, as viewed from the operator seated in an operator's seat in the cabin **10**.

[0033] The lower traveling body **1** includes, for example, a pair of left and right crawlers **1C**. In particular, the crawlers **1C** include a left crawler **1CL** and a right crawler **1CR**. The left crawler **1CL** is driven by a left traveling hydraulic motor **2ML**, and the right crawler **1CR** is driven by a right traveling hydraulic motor **2MR**. The left traveling hydraulic motor **2ML** is a traveling driving part that drives the left crawler **1CL** as a driven part, and can rotate the left crawler **1CL**. The right traveling hydraulic motor **2MR** is a traveling driving part that drives the right crawler **1CR** as a driven part, and can rotate the right crawler **1CR**. The travel driving part may be an electric motor.

[0034] The upper slewing body **3** slews with respect to the lower traveling body **1** by the slewing mechanism **2** being driven by the slewing hydraulic motor **2A**. The slewing hydraulic motor **2A** is a slewing driving unit that drives the upper slewing body **3** as a driven part, and can change the direction of the upper slewing body **3**. The slewing driving part may be an electric motor.

[0035] A boom **4** is rotatably attached to the center of the front portion of the upper slewing body **3**, an arm **5** is rotatably attached to the distal end of the boom **4**, and a bucket **6** is rotatably attached to the distal end of the arm **5**. In the illustrated example, the boom **4**, the arm **5**, and the bucket **6** constitute an excavation attachment which is an example of the attachment **AT**. The boom **4**, the arm **5**, and the bucket **6** are driven by a boom cylinder **7**, an arm cylinder **8**, and a bucket cylinder **9**, respectively.

[0036] The bucket **6** is an example of a work tool (end attachment). The bucket **6** is used for, for example, excavation work. Instead of the bucket **6**, another work tool may be attached to the distal end of the arm **5** depending on the work content or the like. The other work tool may be another type of bucket, such as a large bucket, a slope bucket, or a dredging bucket. The other work tool may be a work tool of a type other than a bucket, such as a stirrer, a breaker, a grapple, or a lifting magnet.

[0037] The slewing hydraulic motor **2A**, the left traveling hydraulic motor **2ML**, the right traveling hydraulic motor **2MR**, the boom cylinder **7**, the arm cylinder **8**, and the bucket cylinder **9** are hydraulic actuators driven by hydraulic fluid discharged from hydraulic pumps.

[0038] In the work machine **100**, all or some of the driven parts, such as the lower traveling body **1**, the upper slewing body **3**, the boom **4**, the arm **5**, and the bucket **6** may be electrically driven. That is, the work machine **100** may be a hybrid work machine, an electric work machine, or the like in which all or some of the driven parts are driven by electric actuators.

[0039] Further, the information transmission device **G1**, an external sound collection device **M1**, an imaging device **S6**, and an external sound output device **SP1** are attached to the work machine **100**.

[0040] The imaging device **S6** is provided on the upper slewing body **3** or the cabin **10**, images the surroundings of the work machine **100**, and acquires image information representing the surroundings of the work machine **100**. In the illustrated example, the imaging device **S6** includes a front camera **S6F**, a left camera **S6L**, a right camera **S6R**, and a rear camera **S6B**.

[0041] The front camera **S6F** is a camera that images the front side of the work machine **100**, and is attached to the outside of the cabin **10**, such as the roof of the cabin **10** or the side surface of the boom **4**. Note that the front camera **S6F** may be attached to the inside of the cabin **10**, such as the ceiling of the cabin **10**, for example. The left camera **S6L** is a camera that images the left side of the work machine **100**, the right camera **S6R** is a camera that images the right side of the work machine **100**, and the rear camera **S6B** is a camera that images the rear side of the work machine **100**. More specifically, each of the front camera **S6F**, the left camera **S6L**, the right camera **S6R**, and the rear camera **S6B** is a monocular wide-angle camera including an image sensor such as a CCD or a CMOS, and outputs a captured image to the display device **D1**. Information of the image captured by the imaging device **S6** is taken into the controller **30**.

[0042] In the illustrated example, the front camera **S6F** is attached to the roof of the cabin **10**, the left camera **S6L** is attached to the left end of the upper surface of the upper slewing body **3**, the right camera **S6R** is attached to the right end of the upper surface of the upper slewing body **3**, and the rear camera **S6B** is attached to the rear end of the upper surface of the upper slewing body **3**.

[0043] The imaging device **S6** may constitute an object detection device that detects an object around the work machine **100**. The object is, for example, a person, an animal, a vehicle, a construction machine, a building, a hole, or the like. The object detection device may be configured to be able to detect a person and an object other than a person in a distinguishable manner. That is, the object detection device may be configured to function as a human detection device. The object detection device may be configured by a device other than a camera. For example, the object detection device may be a LiDAR. The LiDAR is, for example, a device capable of measuring a distance between a point group of one million or more points within a monitoring range and the LiDAR (laser source). The object detection device may be another device capable of measuring the distance to the object, such as a stereo camera, a range image camera, or a millimeter wave radar. When a millimeter wave radar or the like is used as the object detection device, the object detection device may derive the distance and direction of the object by transmitting a large number of signals (laser light or the like) toward the object and receiving the reflected signals. Alternatively, the object detection device may be a combination of two or more types of devices. For example, the object detection device may be a combination of an imaging device and a LiDAR, a combination of an imaging device and a millimeter wave radar, or a combination of an imaging device and a stereo camera.

[0044] The external sound collection device **M1** is a device that collects external sound and is also called a microphone. In the illustrated example, the external sound collection device **M1** is provided in the upper slewing body **3** or the cabin **10**, converts sound (air vibration) generated around the work machine **100** into mechanical vibration, and converts the mechanical vibration into an electric signal. More specifically, the external sound collection device **M1** is configured to be able to pick up voices uttered by the worker around the work machine **100**, and includes a front microphone **M1F**, a left microphone **M1L**, a right microphone **M1R**, and a rear microphone **M1B**.

[0045] The front microphone **M1F** is a microphone that collects sound generated in front of the work machine **100**, and is attached to the outside of the cabin **10**, such as the roof of the cabin **10** or the side surface of the boom **4**. The front microphone **M1F** may be attached to the inside of the cabin **10**, for example, the ceiling of the cabin **10**. The left microphone **M1L** is a microphone that collects sound generated on the left side of the work machine **100**, the right microphone **M1R** is a microphone that collects sound generated on the right side of the work machine **100**, and the rear microphone **M1B** is a microphone that collects sound generated on the rear side of the work machine **100**. Then, the electric signals generated by the front microphone **M1F**, the left microphone **M1L**, the right microphone **M1R**, and the rear microphone **M1B** are taken into the controller **30**.

[0046] In the illustrated example, the front microphone **M1F** is attached to the roof of the cabin **10**, the left microphone **M1L** is attached to the left end of the upper surface of the upper slewing body **3**, the right microphone **M1R** is attached to the right end of the upper surface of the upper slewing body **3**, and the rear microphone **M1B** is attached to the rear end of the upper surface of the upper slewing body **3**. In this way, the four external sound collection devices **M1** (the front microphone **M1F**, the left microphone **M1L**, the right microphone **M1R**, and the rear microphone **M1B**) are provided at different positions of the upper slewing body **3**. Therefore, the controller **30** can detect the direction in which the sound generation source is located based on the difference in the sound collected by each of the four external sound collection devices **M1** (e.g., the difference in the volume). In addition, when an array microphone is used as the external sound collection device **M1**, for example, a direction in which a sound generation source is present can be detected based on a phase shift or a difference in volume.

[0047] In the illustrated example, the four external sound collection devices **M1** and the four imaging devices **S6** are arranged so as to correspond to each other. More specifically, the front microphone **M1F** is arranged adjacent to the front camera **S6F**, the left microphone **M1L** is arranged adjacent to the left camera **S6L**, the right microphone **M1R** is arranged adjacent to the

right camera S6R, and the rear microphone M1B is arranged adjacent to the rear camera S6B.
[0048] The external sound output device SP1 is a device that outputs sound toward the surroundings of the work machine 100. In the illustrated example, the external sound output device SP1 is a non-directional speaker, and is configured to be able to uniformly output sound in all directions. However, the external sound output device SP1 may be a directional speaker that outputs sound in a specific direction such as forward.

[0049] An information transmission device G1 is a device for notifying those on the outside of the work machine 100 of the state of the work machine 100. In the illustrated example, the information transmission device G1 is provided in the upper slewing body 3 or the cabin 10, and is configured to be able to transmit the state of the work machine 100 to the worker around the work machine 100. In particular, the information transmission device G1 is a light emitting device and includes a front light bar G1F, a left light bar G1L, a right light bar G1R, and a rear light bar G1B.

[0050] The front light bar G1F is a light emitting device that can visually convey information to the operator or the like in front of the work machine 100, and is attached to the outside of the cabin 10, such as the roof of the cabin 10 or the side surface of the boom 4. Note that the front light bar G1F may be attached to the inside of the cabin 10, such as the ceiling of the cabin 10, for example. The left light bar G1L is a light emitting device that can visually convey information to a worker or the like on the left side of the work machine 100, the right light bar G1R is a light emitting device that can visually convey information to a worker or the like on the right side of the work machine 100, and the rear light bar G1B is a light emitting device that can visually convey information to a worker or the like on the rear side of the work machine 100. Each of the front light bar G1F, the left light bar G1L, the right light bar G1R, and the rear light bar G1B emits light in response to an electric signal from the controller 30. In the illustrated example, the light emitting device is an LED light, but may be another light emitting device such as a halogen lamp. The light emitting device is a multicolor light emitting type, but may be a monochromatic light emitting type.

[0051] In the illustrated example, the front light bar G1F is attached to the roof of the cabin 10, the left light bar G1L is attached to the left end of the upper surface of the upper slewing body 3, the right light bar G1R is attached to the right end of the upper surface of the upper slewing body 3, and the rear light bar G1B is attached to the rear end of the upper surface of the upper slewing body 3. In this way, the four information transmission devices G1 (the front light bar G1F, the left light bar G1L, the right light bar G1R, and the rear light bar G1B) are provided at different positions of the upper slewing body 3. Therefore, the controller 30 can inform the workers or the like in the front, left, right, and rear of the work machine 100 of the state of the work machine 100 by activating the four information transmission devices G1 separately.

[0052] In the illustrated example, the four information transmission devices G1 and the four external sound collection devices M1 are arranged so as to correspond to each other. In particular, the front light bar G1F is arranged adjacent to the front microphone M1F, the left light bar G1L is arranged adjacent to the left microphone M1L, the right light bar G1R is arranged adjacent to the right microphone M1R, and the rear light bar G1B is arranged adjacent to the rear microphone M1B.

[0053] FIG. 3 is a view illustrating a configuration example of the external sound collection device M1 and the information transmission device G1 attached to the work machine 100. More specifically, FIG. 3 is a perspective view of the left microphone M1L and the left light bar G1L attached to a housing having a substantially rectangular parallelepiped shape. Note that the following description with reference to FIG. 3 relates to a combination of the left microphone M1L and the left light bar G1L, but is similarly applied to a combination of the front microphone M1F and the front light bar G1F, a combination of the right microphone M1R and the right light bar G1R, and a combination of the rear microphone M1B and the rear light bar G1B.

[0054] As illustrated in FIG. 3, the left microphone M1L and the left light bar G1L are disposed on the left side surface of the housing having a substantially rectangular parallelepiped shape so as to

face the left side of the work machine **100**. With this arrangement, the left microphone **M1L** can efficiently collect sound generated on the left side of the work machine **100**, and the left light bar **G1L** can efficiently convey the state of the work machine **100** to the worker on the left side of the work machine **100**. For example, the left microphone **M1L** can capture a voice uttered by a worker on the left side of the work machine **100**, and the left light bar **G1L** can notify the worker that the left microphone **M1L** has captured the voice of the worker by emitting light in a predetermined color. In this case, the worker on the left side of the work machine **100** who speaks to the left microphone **M1L** can confirm that his/her voice has reached the left microphone **M1L** (i.e., the operator of the work machine **100**) by viewing the left light bar **G1L** that emits light in a predetermined color.

[0055] The information transmission device **G1** may be provided on an upper portion of each of the four side surfaces of the cabin **10** (see FIG. **6**). For example, the information transmission device **G1** may be configured such that the front light bar **G1F** is attached to the upper portion of the front surface of the cabin **10**, the left light bar **G1L** is attached to the upper portion of the left surface of the cabin **10**, the right light bar **G1R** is attached to the upper portion of the right surface of the cabin **10**, and the rear light bar **G1B** is attached to the upper portion of the rear surface of the cabin **10**. The information transmission device **G1** may be a single rotating light, such as a Nico Torch (Registered Trademark) attached to the upper surface of the cabin **10**, or may be a display such as a liquid-crystal display or an organic EL display.

[0056] The controller **30** is an example of a control device, and is configured by a computer including a CPU, a volatile storage device, a nonvolatile storage device, various input/output interfaces, and the like, for example. The controller **30** reads the program from the nonvolatile storage device, loads a program into the volatile storage device, and causes the CPU to execute the program, thereby implementing various functions. In the illustrated example, the controller **30** is configured to be able to control the work machine **100** by implementing various functions. The various functions include, for example, a machine guidance function for guiding the manual operation of the work machine **100** by the operator. The various functions may include a contact avoidance function of automatically or autonomously activating or stopping the work machine **100** in order to avoid contact between the work machine **100** and an object present in the monitoring range around the work machine **100**.

[0057] A boom angle sensor **S1** detects a boom angle which is a rotation angle of the boom **4** with respect to the upper slewing body **3**. An arm angle sensor **S2** detects an arm angle which is a rotation angle of the arm **5** with respect to the boom **4**. A bucket angle sensor **S3** detects a bucket angle that is a rotation angle of the bucket **6** with respect to the arm **5**.

[0058] The boom angle sensor **S1**, the arm angle sensor **S2**, and the bucket angle sensor **S3** may be, for example, rotary encoders, accelerometers, six axis sensors, or IMU (Inertial Measurement Unit), or may be potentiometers using variable resistors, or cylinder stroke sensors that detect the stroke amounts of hydraulic cylinders.

[0059] A detection signal corresponding to the boom angle by the boom angle sensor **S1**, a detection signal corresponding to the arm angle by the arm angle sensor **S2**, and a detection signal corresponding to the bucket angle by the bucket angle sensor **S3** are taken into the controller **30**.

[0060] A body inclination sensor **S4** detects an inclination state of the body (the lower traveling body **1** or the upper slewing body **3**) with respect to a horizontal plane.

[0061] The body inclination sensor **S4** is attached to, for example, the upper slewing body **3**, and detects inclination angles of the work machine **100** (i.e., the upper slewing body **3**) around two axes in the front-rear direction and the right-left direction. The body inclination sensor **S4** may be, for example, an accelerometer, a six axis sensor, or an IMU. A detection signal corresponding to the inclination angle by the body inclination sensor **S4** is taken into the controller **30**.

[0062] A slewing sensor **S5** outputs information related to the slewing of the upper slewing body **3**. The slewing sensor **S5** detects, for example, a slewing angular velocity of the upper slewing body **3**

with respect to the lower traveling body **1**. The slewing sensor **S5** may detect a slewing angle. The slewing sensor **S5** may be, for example, a gyro sensor, a resolver, or a rotary encoder. A detection signal corresponding to the slewing angle or slewing angular velocity of the upper slewing body **3** by the slewing sensor **S5** is taken into the controller **30**.

[0063] The positioning device **PS** measures the position of the upper slewing body **3**. The positioning device **PS** is, for example, a global navigation satellite system (GNSS) compass, and detects the position and orientation of the upper slewing body **3**. A detection signal corresponding to the position and orientation of the upper slewing body **3** is taken into the controller **30**. The function of detecting the orientation of the upper slewing body **3** may be implemented by an orientation sensor attached to the upper slewing body **3**.

[0064] The cabin **10** is a partitioned space where an operator gets into and is provided on the front left side of the upper slewing body **3**. However, in a case where the work machine **100** is remotely controlled, or in a case where the work machine **100** is activated by fully automatic driving, the cabin **10** may be omitted.

[0065] A communication device **T1** communicates with external devices via communication networks including mobile communication networks, satellite communication networks, Internet networks, and the like. The communication device **T1** is, for example, a mobile communication module corresponding to a mobile communication standard such as Long Term Evolution (LTE), 4G (4th Generation), or 5G (5th Generation), a communication module corresponding to a short-range wireless communication standard such as Wi-Fi (trademark) or Bluetooth (trademark), or a satellite communications module for connecting to a satellite communications network.

[0066] The work machine **100** activates the actuator in response to an operation of an operator who has entered the cabin **10**, and drives the driven parts such as the lower traveling body **1**, the upper slewing body **3**, the boom **4**, the arm **5**, and the bucket **6**.

[0067] Alternatively, the work machine **100** may be configured to be remotely controlled from the outside of the work machine **100**. When the work machine **100** is remotely controlled, the inside of the cabin **10** may be unmanned.

[0068] The work machine **100** may automatically activate the actuator independent of the content of the operation by the operator. Thus, the work machine **100** implements a function of automatically activating at least some of the driven parts such as the lower traveling body **1**, the upper slewing body **3**, the boom **4**, the arm **5**, and the bucket **6**, that is, a so-called “machine control function”.

[0069] FIG. **4** is a diagram schematically illustrating a configuration example of the work machine **100**. In FIG. **4**, the mechanical power transmission system, the hydraulic fluid line, the pilot line, and the electric control system are indicated by a double line, a thick solid line, a thick broken line, and a dotted line, respectively.

[0070] The drive system of the work machine **100** includes an engine **11**, the regulator **13**, a main pump **14**, and the control valve unit **17**. The hydraulic drive system of the work machine **100** includes hydraulic actuators such as a slewing hydraulic motor **2A**, the left traveling hydraulic motor **2ML**, the right traveling hydraulic motor **2MR**, the boom cylinder **7**, an arm cylinder **8**, and a bucket cylinder **9**.

[0071] The engine **11** is an example of a power source of the work machine **100**, and is mounted, for example, on a rear portion of the upper slewing body **3**. The power source of the work machine **100** may be a combination of a power source such as a battery or a fuel cell and an electric motor. Specifically, the engine **11** rotates at a constant target rotation speed set in advance under direct or indirect control of the controller **30**, and drives the main pump **14** and a pilot pump **15**. The engine **11** is, for example, a diesel engine using diesel as fuel. The engine **11** may be a gasoline engine, a hydrogen engine, or the like.

[0072] The regulator **13** controls the discharge amount of the main pump **14**. For example, the regulator **13** controls the discharge amount of the main pump **14** by adjusting the angle (tilting

angle) of the swash plate of the main pump **14** in response to a control command from the controller **30**.

[0073] The main pump **14** is mounted, for example, on the rear portion of the upper slewing body **3** similarly to the engine **11**, and supplies the hydraulic fluid to the control valve unit **17** through the hydraulic fluid line. In the illustrated example, the main pump **14** is a variable displacement hydraulic pump.

[0074] The control valve unit **17** is one of the hydraulic control devices that control the hydraulic system in the work machine **100**. In the illustrated example, the control valve unit **17** includes control valves **171** to **176**. The control valve unit **17** is configured to be able to selectively supply the hydraulic fluid discharged by the main pump **14** to one or a plurality of hydraulic actuators through the control valves **171** to **176**. The control valves **171** to **176** control the flow rate of the hydraulic fluid flowing from the main pump **14** to the hydraulic actuator and the flow rate of the hydraulic fluid flowing from the hydraulic actuator to the hydraulic fluid tank. The hydraulic actuators include the boom cylinder **7**, an arm cylinder **8**, a bucket cylinder **9**, the left traveling hydraulic motor **2ML**, the right traveling hydraulic motor **2MR**, and the slewing hydraulic motor **2A**. More specifically, the control valve **171** corresponds to the left traveling hydraulic motor **2ML**, the control valve **172** corresponds to the right traveling hydraulic motor **2MR**, and the control valve **173** corresponds to the slewing hydraulic motor **2A**. The control valve **174** corresponds to the bucket cylinder **9**, the control valve **175** corresponds to the boom cylinder **7**, and the control valve **176** corresponds to the arm cylinder **8**.

[0075] The pilot pump **15** is an example of a pilot pressure generating device and is configured to be able to supply the hydraulic fluid to the hydraulic control device via a pilot line. In the illustrated example, the pilot pump **15** is a fixed displacement hydraulic pump. However, the pilot pressure generating device may be implemented by the main pump **14**. That is, the main pump **14** may have a function of supplying the hydraulic fluid to various hydraulic control devices via the pilot line, in addition to the function of supplying the hydraulic fluid to the control valve unit **17** via the hydraulic fluid line. In this case, the pilot pump **15** may be omitted.

[0076] The discharge pressure sensor **28** is configured to detect the discharge pressure of the main pump **14**. In the illustrated example, the discharge pressure sensor **28** outputs the detected value to the controller **30**.

[0077] The operation device **26** is a device used by an operator to operate the actuator. The operation device **26** includes, for example, an operation lever and an operation pedal. The actuator may be a hydraulic actuator or an electric actuator.

[0078] The operation sensor **29** is configured to detect the content of an operation performed by the operator using the operation device **26**. In the present embodiment, the operation sensor **29** detects the operation direction and the operation amount of the operation device **26** corresponding to each of the actuators, and outputs the detected values to the controller **30**. In the illustrated example, the controller **30** can control the opening area of a valve **31** in accordance with the output of the operation sensor **29**. The controller **30** supplies the hydraulic fluid discharged from the pilot pump **15** to the pilot port of the corresponding control valve in the control valve unit **17**. The pressure of the hydraulic fluid supplied to each of the pilot ports (pilot pressure) is, in principle, a pressure corresponding to the operation direction and the operation amount of the operation device **26** corresponding to each of the hydraulic actuators. In this way, the operation device **26** is configured to be able to supply the hydraulic fluid discharged by the pilot pump **15** to the pilot port of the corresponding control valve in the control valve unit **17**.

[0079] The valve **31** functioning as a control valve for machine control is disposed in a conduit connecting the pilot pump **15** and a pilot port of a control valve in the control valve unit **17**, and is configured to be able to change a flow passage area of the conduit. In the illustrated example, the valve **31** is an electromagnetic valve that is activated in response to a control command output from the controller **30**. Therefore, the controller **30** can adjust the pilot pressure acting on the pilot port

of the control valve by the valve **31**, independently of the operation of the operation device **26** by the operator.

[0080] With this configuration, even when an operation is not performed on a specific operation device **26**, the controller **30** can activate the hydraulic actuator corresponding to the specific operation device **26**.

[0081] As illustrated in FIG. **4**, the control system of the work machine **100** includes the controller **30**, the display device **D1**, the input device **D2**, a horn button **HS**, the speech button **KS**, the external sound collection device **M1**, an internal sound collection device **M2**, the external sound output device **SP1**, an internal sound output device **SP2**, the external volume dial **DL1**, the internal volume dial **DL2**, the switch **SW**, the communication device **T1**, and the like.

[0082] The controller **30** is configured to be able to output a control command to the regulator **13** as necessary and change the discharge amount of the main pump **14**. Further, the controller **30** may be configured to perform control related to a machine guidance function for guiding (leading) manual operation of the work machine **100** by the operator through the operation device **26**, for example. Further, the controller **30** may be configured to perform control related to a machine control function of automatically supporting manual operation of the work machine **100** by the operator through the operation device **26**, for example. Note that some of the functions of the controller **30** may be implemented by another controller (control device). That is, the functions of the controller **30** may be implemented in a distributed manner by a plurality of controllers. For example, the machine guidance function and the machine control function may be implemented by a dedicated controller (control device).

[0083] Here, the inside of the cabin **10** will be described with reference to FIG. **5**. FIG. **5** is a top view illustrating an inside of the cabin **10**. The work machine **100** includes an operator's seat **50**, an operation device **26**, a display device **D1**, and the like, which are disposed inside the cabin **10**. A door for getting in and out is provided on the left side of the operator's seat **50**. The operator can enter the inside of the cabin **10** by opening the entrance door.

[0084] The operator's seat **50** is disposed at the center of the cabin **10** in a top view. The operator's seat **50** includes a seat **51** on which an operator sits and a backrest **52**. The operator's seat **50** is a reclining seat, and the inclination angle of the backrest **52** is adjustable. A left arm rest **53L** is disposed on the left side of the operator's seat **50**, and a right arm rest **53R** is disposed on the right side of the operator's seat **50**. The left arm rest **53L** and the right arm rest **53R** are rotatably supported by the backrest **52**.

[0085] A left console **54L** is disposed on the left side of the operator's seat **50**, and a right console **54R** is disposed on the right side of the operator's seat **50**. The left console **54L** and the right console **54R** extend along the front-rear direction. The operator's seat **50** is slidable in the front-rear direction. The operator's seat **50** may be configured to be slidable in the front-rear direction together with the left console **54L** and the right console **54R**.

[0086] The left arm rest **53L** is disposed on the left console **54L**. The right arm rest **53R** is disposed on the right console **54R**. The left arm rest **53L** is disposed so as to cover a part of the left console **54L** in a top view. The right arm rest **53R** is disposed so as to cover a part of the right console **54R** in a top view.

[0087] The operation device **26** includes a left operation lever **26L**, a right operation lever **26R**, a left traveling pedal **26PL**, a right traveling pedal **26PR**, a left traveling lever **26DL**, and a right traveling lever **26DR**. The left operation lever **26L** is provided at a front portion of the left console **54L**. Similarly, the right operation lever **26R** is provided at a front portion of the right console **54R**. The operator seated in the operator's seat **50** can operate the left operation lever **26L** while gripping the left operation lever **26L** with the left hand, and can operate the right operation lever **26R** while gripping the right operation lever **26R** with the right hand. The operator seated in the operator's seat **50** can operate the left operation lever **26L** with the left hand to drive the arm cylinder **8** and the slewing hydraulic motor **2A**. The operator seated in the operator's seat **50** can operate the right

operation lever **26R** with the right hand to drive the boom cylinder **7** and the bucket cylinder **9**. The base of each of the left operation lever **26L** and the right operation lever **26R** is covered with a lever boot **27**.

[0088] The left traveling pedal **26PL** and the right traveling pedal **26PR** are disposed on the floor surface in front of the operator's seat **50**. The operator seated in the operator's seat **50** can operate the left traveling pedal **26PL** with his/her left foot to drive the left traveling hydraulic motor **2ML**. The operator seated in the operator's seat **50** can operate the right traveling pedal **26PR** with the right foot to drive the right traveling hydraulic motor **2MR**.

[0089] The left traveling lever **26DL** and the right traveling lever **26DR** are disposed between the left traveling pedal **26PL** and the right traveling pedal **26PR** in a top view. The left traveling lever **26DL** and the right traveling lever **26DR** extend upward from the floor surface in front of the operator's seat **50**. The operator seated in the operator's seat **50** can drive the left traveling hydraulic motor **2ML** by operating the left traveling lever **26DL** while gripping the left traveling lever **26DL** with the left hand, similarly to the operation via the left traveling pedal **26PL**. Further, the operator seated in the operator's seat **50** can drive the right traveling hydraulic motor **2MR** by operating the right traveling lever **26DR** while gripping the right traveling lever **26DR** with the right hand, similarly to the operation via the right traveling pedal **26PR**. The left traveling lever **26DL** and the right traveling lever **26DR** are arranged so that the operator can operate the left traveling lever **26DL** and the right traveling lever **26DR** simultaneously with one hand.

[0090] The display device **D1** is provided at a position in the cabin **10** where the display device **D1** is easily visible to the seated operator, and displays various information images under the control of the controller **30**. In the illustrated example, the display device **D1** is disposed on the right front side of the operator's seat **50** and is connected to the controller **30** via a dedicated line. The display device **D1** displays various kinds of image information. The display device **D1** includes a display screen that displays information such as the work conditions or the activation state of the work machine **100**. The operator seated in the operator's seat **50** can perform work with the work machine **100** while checking various information displayed on the display device **D1**. The display device **D1** may be provided with an input device **D2**.

[0091] The input device **D2** is provided within reach of the operator seated in the operator's seat **50**, receives various operation inputs from the operator, and outputs signals corresponding to the operation inputs to the controller **30**. The input device **D2** includes a touch panel mounted on a display of the display device **D1** that displays various information images, a knob switch provided at a tip of one or more lever portions of a plurality of operation levers included in the operation device **26**, or a button switch, a lever, a toggle switch, a rotary dial, or the like installed around the display device **D1**. A signal corresponding to the content of the operation on the input device **D2** is taken into the controller **30**.

[0092] A gate bar **55** is attached to the front surface of the front end portion of the left console **54L**. The gate bar **55** activates in conjunction with an operation state of a gate lock lever **GL** provided on the left console **54L**. The gate bar **55** is attached to a frame inside the left console **54L** so as to be able to rise and fall about a shaft extending in the left-right direction at the upper end portion.

[0093] The gate lock lever **GL** is a mechanical input operation unit for switching between a state in which the work machine **100** can be operated by the operation device **26** (operable state) and a state in which the work machine **100** cannot be operated by the operation device **26** (inoperable state). In the illustrated example, the gate lock lever **GL** is configured such that the operator can switch between a first operation position for implementing the inoperable state and a second operation position for implementing the operable state. The controller **30** switches between an operable state and an inoperable state in accordance with the operation state of the gate lock lever **GL**. In the illustrated example, the controller **30** electrically switches the pilot line between a communication state and a non-communication state in accordance with the operation state of the gate lock lever **GL**, thereby switching the work machine **100** between the operable state and the inoperable state.

[0094] When the gate lock lever GL is at the second operation position, the gate bar **55** is in a state of being raised forward so as to prevent the operator from passing through the entrance door (passing prohibited state), as illustrated in FIG. 5. When the gate lock lever GL is at the first operation position, the gate bar **55** is housed in the left console **54L** so as not to hinder the operator from passing through the entrance door (passing permission state).

[0095] With this configuration, the operator cannot operate the work machine **100** unless the gate lock lever GL is located at the second operation position to set the gate bar **55** in the passing prohibited state. Therefore, this configuration can prevent the work machine **100** from moving unintentionally even when the operator touches the operation device **26** inadvertently when getting in and out. Therefore, this configuration can improve the safety of the work machine **100**.

[0096] The work machine **100** may be configured to be able to receive a predetermined operation for starting the engine **11** only when the gate lock lever GL is at the second operation position and the gate bar **55** is in the passing prohibited state. That is, the work machine **100** may be configured such that the engine **11** cannot be started when the gate lock lever GL is at the first operation position and the gate bar **55** is in the passing permission state.

[0097] A switch SW is provided on the right console **54R**. A window side console **56** is installed on the right side of the right console **54R**. The window side console **56** extends over the entire length of the cabin **10** in the front-rear direction, and is provided so as to be parallel to the right console **54R**. The display device **D1** is installed in front of the window side console **56**. The window side console **56** is provided with an external volume dial **DL1**, an internal volume dial **DL2**, an internal sound collection device **M2**, a radio tuner, and the like. The radio tuner or the like may be installed on the left console **54L** or the right console **54R**.

[0098] An internal sound collection device **M2** is a device that collects sound generated in the cabin **10**. In the illustrated example, the internal sound collection device **M2** is an indoor microphone, and is configured to be able to pick up voices uttered by the operator in the cabin **10**.

[0099] The horn button **HS** is a button operated by the operator of the work machine **100** when sounding a horn sound. In the illustrated example, the horn button **HS** is a knob switch provided at the distal end of the left operation lever **26L**.

[0100] The speech button **KS** is a button operated by the operator of the work machine **100** when the operator of the work machine **100** speaks to a worker around the work machine **100**. In the illustrated example, the speech button **KS** is a knob switch provided at the tip of the right operation lever **26R**.

[0101] The internal sound output device **SP2** is a device that outputs sound toward the operator in the cabin **10**, and is provided in the cabin **10**. The internal sound output device **SP2** converts an electric signal input from the controller **30** into a physical sound (vibration of air) and outputs the physical sound. The internal sound output device **SP2** may be provided at any position, and may be provided, for example, in the vicinity of the display device **D1**, in the vicinity of the input device **D2**, or in the vicinity of the entrance door of the cabin **10**. In the illustrated example, the internal sound output device **SP2** includes a left indoor speaker **SP2L** attached to an upper left corner of the rear wall of the cabin **10** and a right indoor speaker **SP2R** attached to an upper right corner of the rear wall of the cabin **10**. The internal sound output device **SP2** may be a headphone or an earphone worn by the operator. In this case, the headphone or earphone is connected to the controller **30** via, for example, Bluetooth (registered trademark) so as to be able to communicate with the controller **30**.

[0102] The external volume dial **DL1** is configured to adjust the volume of the sound output by the external sound output device **SP1**. The volume of the sound output by each external sound output device **SP1** may be additionally adjusted by using a device other than the external volume dial **DL1**, such as a touch panel attached to the display device **D1**.

[0103] The external volume dial **DL1** may be configured to be infinitely rotatable in each of the clockwise and counterclockwise directions. This is to accommodate a case where the volume

adjustment using the external volume dial DL1 and the volume adjustment using a device other than the external volume dial DL1 are used together.

[0104] The internal volume dial DL2 is configured to adjust the volume of the sound output by the internal sound output device SP2. The volume of the sound output from each internal sound output device SP2 may be additionally adjusted by using a device other than the internal volume dial DL2, such as a touch panel attached to the display device D1.

[0105] The internal volume dial DL2 may be configured to be infinitely rotatable in each of the clockwise and counterclockwise directions. This is to accommodate a case where the volume adjustment using the internal volume dial DL2 and the volume adjustment using a device other than the internal volume dial DL2 are used together.

[0106] The switch SW is an example of an operation tool for switching the activation state of the conversation function. In the illustrated example, the switch SW is provided on the upper surface of the right console 54R. However, the switch SW may be one of the input devices D2, may be implemented by a touch panel provided on the display device D1, or may be a knob switch.

[0107] The conversation function is a function for implementing a conversation between the operator OP of the work machine 100 and the worker WK around the work machine 100 as illustrated in FIG. 6. FIG. 6 is a perspective view illustrating the work machine 100 on which the operator OP is riding and the worker WK in front of the work machine 100 on the left. FIG. 6 shows a state in which the voice of the operator OP is being collected by the internal sound collection device M2 and output from the external sound output device SP1, and the voice of the worker WK is collected by the external sound collection device M1 and output from the internal sound output device SP2. In the work machine 100 illustrated in FIG. 6, the information transmission device G1 is provided on the upper portion of each of the four side surfaces of the cabin 10. The front light bar G1F provided at the upper portion of the front surface of the cabin 10 emits green light, and the left light bar G1L provided at the upper portion of the left surface of the cabin 10 emits white light. In FIG. 6, a dot pattern is given to the front light bar G1F emitting green light. The worker WK who sees the front light bar G1F emitting green light can recognize that his/her voice is detected by the front microphone M1F. Note that, in FIG. 6, other devices such as the imaging device S6 are not illustrated for the sake of clarity.

[0108] The activation state of the conversation function includes an ON state (state illustrated in FIG. 6) in which a conversation between the operator OP and the worker WK is possible, and an OFF state in which a conversation between the operator OP and the worker WK is not possible. However, the activation state of the conversation function may additionally include at least one of a hearable state (related to the operator OP) in which the operator OP can hear the voice of the worker WK but the worker WK cannot hear the voice of the operator OP and an utterable state (related to the operator OP) in which the worker WK can hear the voice of the operator OP but the operator OP cannot hear the voice of the worker WK.

[0109] In detail, when the activation state of the conversation function is switched to the ON state by operating the switch SW, the external sound collection device M1, the external sound output device SP1, the internal sound collection device M2, and the internal sound output device SP2 are in an available state. When the switch SW is operated and the activation state of the conversation function is switched to the OFF state, the external sound collection device M1, the external sound output device SP1, the internal sound collection device M2, and the internal sound output device SP2 are in the unavailable state. When the switch SW is operated to switch the activation state of the conversation function to the hearable state, the external sound collection device M1 and the internal sound output device SP2 are brought into the available state. When the switch SW is operated to switch the activation state of the conversation function to the utterable state, the external sound output device SP1 and the internal sound collection device M2 are enabled. In the illustrated example, the operator OP can speak to the worker WK using the external sound output device SP1 by speaking while pressing the speech button KS when the internal sound collection

device M2 is in an available state.

[0110] Next, referring to FIG. 7, a process (hereinafter, referred to as an “information transmission process”) in which the controller 30 controls the information transmission device G1 to transmit information to the surroundings of the work machine 100 will be described. FIG. 7 is a flowchart illustrating an example of a sequence of an information transmission process. The controller 30 repeatedly executes the information transmission process at a predetermined control cycle.

[0111] First, the controller 30 determines whether or not the conversation function is in the ON state (step ST1). In the illustrated example, the operator OP can switch the activation state of the conversation function between the ON state and the OFF state by operating the switch SW.

[0112] When the controller 30 determines that the conversation function is in the ON state (YES in step ST1), the controller 30 determines whether or not the external sound collection device M1 has detected a human voice (step ST2). In the illustrated example, the controller 30 analyzes frequencies of electric signals corresponding to sounds collected by the external sound collection device M1, and determines whether or not a voice of a person (the worker WK) is included in the sounds collected by the external sound collection device M1, that is, whether or not the external sound collection device M1 has picked up a voice of a person. The controller 30 may determine whether or not a human voice is included based on waveform patterns of electric signals corresponding to the sounds collected by the external sound collection device M1. Further, the functions related to these sounds may be implemented by a device (e.g., an independent sound controller) different from the controller 30.

[0113] When the controller 30 determines that the external sound collection device M1 has detected a human voice (YES in step ST2), the controller 30 causes the information transmission device G1 to emit light in the first mode (step ST3). In the illustrated example, the controller 30 outputs a control command to each of the four information transmission devices G1 (the rear light bar G1B, the front light bar G1F, the left light bar G1L, and the right light bar G1R), and causes each of the four information transmission devices G1 to emit green light.

[0114] The worker WK who sees the information transmission device G1 emitting green light due to the utterance of the worker WK can recognize that the voice of the worker WK has reached the operator OP of the work machine 100. This is because the worker WK can assume that his/her voice collected by the external sound collection device M1 should have been output from the internal sound output device SP2.

[0115] When the controller 30 determines that the external sound collection device M1 does not detect a human voice (NO in step ST2), the controller 30 determines whether the internal sound collection device M2 has detected a human voice (step ST4). In the illustrated example, the controller 30 analyzes frequencies of electric signals corresponding to sounds collected by the internal sound collection device M2, and determines whether or not a voice of a person (operator OP) are included in the sounds collected by the internal sound collection device M2, that is, whether or not the internal sound collection device M2 has picked up the voice of the person. The controller 30 may determine whether or not a human voice is included based on a waveform pattern of an electric signal corresponding to the sound collected by the internal sound collection device M2.

[0116] When the controller 30 determines that the internal sound collection device M2 has detected a human voice (YES in step ST4), the controller 30 causes the information transmission device G1 to emit light in the second mode (step ST5). In the illustrated example, the controller 30 outputs a control command to each of the four information transmission devices G1, and causes each of the four information transmission devices G1 to emit blue light. The worker WK who sees the information transmission device G1 emitting blue light can recognize that the operator OP of the work machine 100 is speaking toward the outside.

[0117] When the controller 30 determines that the internal sound collection device M2 does not detect a human voice (NO in step ST4), the controller 30 causes the information transmission

device G1 to emit light in the third mode (step ST6). In the illustrated example, the controller 30 outputs a control command to each of the four information transmission devices G1, and causes each of the four information transmission devices G1 to emit white light. The worker WK who sees the white light emitted by the information transmission device G1 can recognize that the conversation function of the work machine 100 is in the ON state. Further, the worker WK can recognize that the operator OP of the work machine 100 is not speaking. That is, the worker WK can recognize that it is a timing suitable for the worker WK to speak.

[0118] In addition, when the information transmission device G1 emits light in white instead of green even though the worker WK is speaking, the worker WK can recognize that his/her voice is not detected by the external sound collection device M1 for a certain factor. The certain factor is, for example, the loudness of the voice of the worker WK, the length between the work machine 100 (external sound collection device M1) and the worker WK, or the loudness of surrounding noise.

[0119] When the controller 30 determines that the conversation function is not in the ON state (NO in step ST1), the controller 30 determines whether or not the external sound collection device M1 has detected a human voice (step ST7).

[0120] Then, when the controller 30 determines that the external sound collection device M1 has detected a human voice (YES in step ST7), the controller 30 causes the information transmission device G1 to emit light in the fourth mode (step ST8). In the illustrated example, the controller 30 outputs a control command to each of the four information transmission devices G1, and causes each of the four information transmission devices G1 to emit red light. The worker WK who sees the information transmission device G1 emitting red light due to the utterance of the worker WK can recognize that the external sound collection device M1 is in an available state but the voice of the worker WK does not reach the operator OP of the work machine 100. This is because the internal sound output device SP2 is not in an available state, and thus the worker WK can assume that his/her voice collected by the external sound collection device M1 is not output from the internal sound output device SP2.

[0121] When the controller 30 determines that the external sound collection device M1 does not detect a human voice (NO in step ST7), the controller 30 turns off the information transmission device G1 (step ST9). In the illustrated example, the controller 30 outputs a control command to each of the four information transmission devices G1 to stop the light emission of each of the four information transmission devices G1. Each of the four information transmission devices G1 stops light emission when a corresponding one of the four information transmission devices G1 emits light, and maintains the state as it is when the corresponding one of the four information transmission devices G1 does not emit light.

[0122] The worker WK who sees the information transmission device G1 not emitting light can recognize that the conversation function of the work machine 100 is not in the ON state. That is, even if the worker WK speaks to the external sound collection device M1, the worker WK can recognize that his/her voice is not transmitted to the operator OP.

[0123] In this way, the worker WK can recognize the light emission mode of the information transmission device G1 to grasp the current situation such as whether or not the voice of the worker WK is being reached the operator OP or whether or not the operator OP is speaking.

[0124] In the above-described example, the first to fourth modes are distinguished from each other by the difference in the emission color. However, the first to fourth modes may be distinguished from each other by the difference in the shade, the luminance, the blinking pattern, or the like, or a combination thereof. For example, each of the first to fourth modes may be implemented by changing the blinking pattern of a light emitter that emits light in a single color. The changing of the blinking pattern may be implemented by changing at least one of the turn-on time and the turn-off time of the light emitter.

[0125] In addition, the emission color used in each of the first to fourth modes distinguished by the

difference in emission color is desirably an emission color (e.g., blue) other than the emission colors (green, yellow, and red) used in the three color lamp (revolving light) attached to the hydraulic excavator with a crane function. This is to avoid the worker WK confusing the luminescent color of the information transmission device G1 indicating the activation state of the conversation function and the luminescent color of the three color lamp indicating the load state of the hydraulic excavator with a crane function. Further, the emission color used in each of the first to fourth modes is desirably an emission color (e.g., white) other than the emission color (e.g., blue) of the light emitter attached to the remote-control type work machine. This is to avoid the worker WK confusing the luminescent color of the information transmission device G1 indicating the activation state of the conversation function and the luminescent color of the light emitter indicating the remote-control type work machine.

[0126] In the above example, the controller 30 is configured to cause all of the four information transmission devices G1 to emit light, but may be configured to cause some of the four information transmission devices G1 to emit light. For example, the controller 30 may be configured to cause one of the four light sources that is located closest to the worker WK who utters a voice to emit light and not to cause the remaining three light sources to emit light. In this case, the controller 30 may determine which of the four information transmission devices G1 is located closest to the worker WK based on a difference in detection timing of the voice of the worker WK, a difference in volume of the detected voice, or the like. Alternatively, the controller 30 may be configured to cause two of the light sources located relatively close to the worker WK who utters voices to emit light and not to cause the remaining two light sources located relatively far from the worker WK to emit light.

[0127] In the above-described example, the information transmission device G1 is configured to include the light emitter (light bar) attached to the work machine 100, but the light emitter (light bar) may be installed in the work site instead of being attached to the work machine 100. In this case, the information transmission device G1 may be a transmission device that transmits a signal to a light emitter (light bar) installed in the work site. In this configuration, the worker WK can recognize the current situation, for example, whether or not his/her voice has reached the operator OP or whether or not the operator OP is speaking, by recognizing the light emission mode of the light emitter (light bar) installed in the work site.

[0128] Alternatively, when the information transmission device G1 is a transmission device, the light emitter (light bar) may be replaced with a mobile device such as a smartphone carried by the worker WK. In this case, the mobile device that has received a signal from the information transmission device G1 serving as the transmission device may cause a light emitting device such as a display screen to emit light in a predetermined mode, may cause the mobile device to vibrate in a predetermined mode, or may cause a sound output device such as a speaker or a buzzer attached to the mobile device to output sound in a predetermined mode. In this configuration, the worker WK can recognize the current situation such as whether or not his/her voice has reached the operator OP or whether or not the operator OP is speaking by recognizing the light emission mode, the vibration mode, or the sound output mode of the light emitting device of the mobile terminal carried by the worker WK.

[0129] Next, with reference to FIG. 8, a process (hereinafter, referred to as a “sound transmission process”) in which the controller 30 transmits the sound around the work machine 100 to the operator OP will be described. FIG. 8 is a flowchart illustrating an example of a proceed of the sound transmission process. The controller 30 repeatedly executes the sound transmission process at a predetermined control cycle.

[0130] First, the controller 30 determines whether the external sound collection device M1 has detected a human voice (step ST11). In the illustrated example, the controller 30 analyzes the frequencies of the electric signals corresponding to the sounds collected by the external sound collection device M1 and determines whether or not the sounds collected by the external sound

collection device **M1** include a voice of a person (worker **WK**), that is, whether or not the external sound collection device **M1** has picked up a voice of a person, as in the information transmission process. The controller **30** may determine whether or not a human voice is included based on a waveform pattern of an electric signal corresponding to a sound collected by the external sound collection device **M1**.

[0131] When the controller **30** determines that the external sound collection device **M1** has detected a human voice (YES in step **ST11**), the controller **30** causes the internal sound output device **SP2** to output a sound in a state in which the volume of a sound other than a human voice is reduced (step **ST12**). In the illustrated example, the sound other than the human voice is, for example, a sound generated by the work machine **100**, a sound generated by another work machine other than the work machine **100**, or the like. The controller **30** adjusts the waveform of the electric signal corresponding to the sound collected by the external sound collection device **M1** so that the volume of the sound other than the voice of the person (worker **WK**) among the sounds collected by the external sound collection device **M1** is reduced, and outputs the sound corresponding to the adjusted electric signal from the internal sound output device **SP2**. The sound other than the human voice may not include a specific sound. The specific sound is, for example, a sound such as a traveling alarm, a whistle, or a warning sound. That is, the controller **30** may cause the internal sound output device **SP2** to output sound so that the volume of human voice and specific sound is not reduced. This is because the sound such as the traveling alarm, the whistle, or the warning sound is considered to be a sound that should be heard by the operator **OP** even during the conversation.

[0132] By this process, the operator **OP** of the work machine **100** can easily hear the voice of the worker **WK** compared to a case where the volume of the sound other than the voice of the person (worker **WK**) is not reduced. Therefore, this configuration can prevent the occurrence of a situation in which the operator **OP** does not notice that the worker **WK** is speaking.

[0133] When the controller **30** determines that the external sound collection device **M1** does not detect a human voice (NO in step **ST11**), the controller **30** causes the internal sound output device **SP2** to output a sound in a state in which the volume of the predetermined sound is reduced (step **ST13**). In the illustrated example, the predetermined sound is, for example, a sound generated by the engine **11** mounted on the work machine **100**. The controller **30** adjusts the waveform of the electric signal corresponding to the sound collected by the external sound collection device **M1** so that the volume of a predetermined sound among the sounds collected by the external sound collection device **M1** is reduced, and outputs the sound corresponding to the adjusted electric signal from the internal sound output device **SP2**. Specifically, the controller **30** can adjust the waveform of the electric signal so that the volume of the predetermined sound is reduced by analyzing at least one of the frequency and the waveform pattern of the electric signals corresponding to the sounds collected by the external sound collection device **M1**. More specifically, the controller **30** can adjust the waveform of the electric signal so that the volume of the predetermined sound is reduced by using at least one of the frequency and the waveform pattern of the predetermined sound stored in advance in the nonvolatile storage device or the like.

[0134] By this processing, the operator **OP** of the work machine **100** can easily hear sounds other than the predetermined sound (e.g., sounds such as a traveling alarm, a whistle, or a warning sound), compared to a case where the volume of the predetermined sound is not suppressed. Therefore, this configuration can prevent the operator **OP** from failing to hear sounds other than the predetermined sound.

[0135] Next, a process of notifying the worker **WK** that the operator **OP** is attempting to have a conversation (hereinafter, referred to as a “conversation notification process”) will be described with reference to FIG. 9. FIG. 9 is a flowchart illustrating an example of a proceed of the conversation notification process. The controller **30** repeatedly executes the conversation notification process at a predetermined control cycle.

[0136] First, the controller **30** determines whether the operator OP is attempting to have a conversation (step ST21). In the illustrated example, when the speech button KS is pressed while the conversation function is in an ON state, the controller **30** determines that the operator OP is attempting to have a conversation. However, the controller **30** may determine that the operator OP is attempting to have a conversation when determining that the internal sound collection device M2 has detected a human voice independent of whether the speech button KS is pressed, or may determine that the operator OP is attempting to have a conversation when determining that the internal sound collection device M2 has detected a human voice in a state where the speech button KS is pressed. Alternatively, the controller **30** may determine that the operator OP is attempting to have a conversation when the internal sound collection device M2 has detected that a specific word is uttered in the cabin **10**, independent of whether the speech button KS is pressed. The specific word is, for example, “conversation start” or “conversation function”.

[0137] When the controller **30** determines that the operator OP is not attempting to have a conversation (NO in step ST21), the controller **30** ends the conversation notification process of this time without giving any notification to the surroundings of the work machine **100**.

[0138] When the controller **30** determines that the operator OP is attempting to have a conversation (YES in step ST21), the controller **30** determines whether or not it is necessary to notify those on the outside of the fact (that the operator OP is attempting to have a conversation) (step ST22). In the illustrated example, the controller **30** determines that it is necessary to notify those on the outside that the operator OP is attempting to have a conversation when the elapsed time from the time point when the previous notification is performed exceeds a predetermined time (e.g., one minute). In other words, when the elapsed time from the time point at which the previous notification is performed is shorter than a predetermined time (e.g., one minute), the controller **30** determines that it is not necessary to notify those on the outside that the operator OP is attempting to have a conversation.

[0139] In the illustrated example, the notification to the outside is implemented by outputting a predetermined sound from the external sound output device SP1. The predetermined sound is a sound for informing the surroundings that the operator OP is attempting to have a conversation. The predetermined sound may be, for example, a relatively high electronic sound such as “beep”, or may be a sound that does not drown out a human voice even when output simultaneously with the human voice. Further, the predetermined sound is preferably output in a mode different from that of the horn sound so that the worker WK does not confuse the horn sound and the predetermined sound. In the illustrated example, the horn sound is a sound output from the horn device when the horn button HS is pressed. Specifically, the predetermined sound is output so as to be different from the horn sound in at least one of sound pressure, tone, and frequency. The notification to the outside may be implemented by the horn device as long as the horn device can output a predetermined sound that is a sound different from the horn sound. The notification to the outside may be implemented by activating the information transmission device G1. In particular, the notification to the outside may be implemented by causing the information transmission device G1 to emit light. Alternatively, in a case where the information transmission device G1 is a transmission device, the notification to the outside may be implemented by activating a mobile device such as a smartphone carried by the worker WK. Specifically, the notification to the outside may be implemented by causing the light emitting device such as a display screen of the mobile terminal to emit light in a predetermined manner, causing the mobile terminal to vibrate in a predetermined manner, or causing a sound output device such as a speaker or a buzzer attached to the mobile terminal to output sound in a predetermined manner. In this configuration, the worker WK can recognize that the operator OP is attempting to have a conversation by recognizing, for example, the light emission mode, the vibration mode, or the sound output mode of the light emitting device of the mobile terminal carried by the worker WK. In this way, the notification to the outside may be a visual notification, a tactile notification, or an auditory notification.

[0140] The elapsed time from the time point when the previous notification is performed is, for example, an elapsed time from the time point when the external sound output device SP1 outputs the latest predetermined sound. However, the elapsed time from the time point when the previous notification is performed may be the elapsed time from the time point when the external sound collection device M1 stops collecting a voice of a person (worker WK), the elapsed time from the time point when the internal sound collection device M2 stops collecting a voice of a person (operator OP), the elapsed time from the time point when the operator OP stops pressing the speech button KS, or the like.

[0141] When the controller 30 determines that it is not necessary to notify those on the outside that the operator OP is attempting to have a conversation (NO in step ST22), the controller 30 ends the conversation notification process of this time without giving any notification to the surroundings of the work machine 100. This is because the worker WK is considered to be still concentrating his/her attention on his/her ear when the worker WK has been having a conversation with the operator OP for the last time. This is also because it is considered that the worker WK feels annoyed if the notification is frequently performed in the middle of the conversation.

[0142] When the controller 30 determines that it is necessary to notify those on the outside that the operator OP is attempting to have a conversation (YES in step ST22), the controller 30 notifies those on the outside of the above fact (step ST23). In the illustrated example, the controller 30 outputs a control command to the external sound output device SP1, and causes the external sound output device SP1 to output a relatively high electronic sound such as “beep” for a predetermined time. The predetermined time is, for example, less than one second.

[0143] The worker WK who hears the predetermined sound can recognize that the operator OP is attempting to have a conversation. Therefore, the worker WK can concentrate his/her attention on his/her ear before the operator OP speaks, and can avoid missing the voice of the operator OP.

[0144] In the illustrated example, when the external sound output device SP1 outputs a predetermined sound, the controller 30 outputs a control command to each of the external sound collection device M1 and the internal sound collection device M2 so that the external sound collection device M1 does not collect a sound and the internal sound collection device M2 does not collect a sound. This is to reduce the occurrence of howling. However, the external sound collection device M1 may be configured to selectively not collect the predetermined sound while collecting sounds other than the predetermined sound when the external sound output device SP1 outputs the predetermined sound, or may be configured to remove a portion corresponding to the predetermined sound from the electric signal generated based on the collected sound. The same applies to the internal sound output device SP2.

[0145] With this configuration, the worker WK can know that the operator OP is attempting to have a conversation before the operator OP of the work machine 100 starts talking by hearing the predetermined sound. Therefore, the worker WK can smoothly proceed with the subsequent conversation without missing the voice of the operator OP at the start of the conversation. Even in a situation where a plurality of work machines are being activated in the same work site, the worker WK can easily recognize which operator of the work machine is attempting to have a conversation.

[0146] In the above-described example, the controller 30 is configured to be able to notify the worker WK that the operator OP of the work machine 100 is attempting to have a conversation with the worker WK around the work machine 100. However, the controller 30 may be configured to notify a person who is away from the work site that the operator OP is attempting to have a conversation with the person. For example, the controller 30 may be configured to notify the manager that the operator OP is attempting to have a conversation with the manager at a management center located away from the work site.

[0147] In the above example, the controller 30 is configured to notify those on the outside of the fact that the operator is attempting to have a conversation when the controller 30 determines that it is necessary to notify those on the outside of the above fact.

[0148] Next, another configuration example of a work machine **100** will be described with reference to FIG. **10**. FIG. **10** is a top view illustrating another configuration example of the work machine **100**. The work machine **100** illustrated in FIG. **10** is different from the work machine **100** illustrated in FIG. **1** in that the external sound output device SP1 is constituted by four speakers (a front speaker SP1F, a left speaker SP1L, a right speaker SPR, and a rear speaker SP1B). In the work machine **100** illustrated in FIG. **1**, the external sound output device SP1 is configured by one omnidirectional speaker provided on the cabin **10**.

[0149] With this configuration, the work machine **100** illustrated in FIG. **10** can output sound toward the worker WK in front of the work machine **100** without outputting sound toward workers WK on the left, right, and rear of the work machine **100** by turning on the front speaker SP1F (a state in which sound can be output) and turning off the left speaker SP1L, the right speaker SPR, and the rear speaker SP1B (a state in which sound cannot be output), for example.

[0150] In the example illustrated in FIG. **10**, the front camera S6F and the front microphone M1F are provided adjacent to the front speaker SP1F, and a front light bar G1F is provided in the housing of the front microphone M1F. Further, a left camera S6L and a left microphone M1L are provided adjacent to the left speaker SP1L, and a left light bar G1L is provided in a housing of the left microphone M1L. Further, a right camera S6R and a right microphone M1R are provided adjacent to the right speaker SP1R, and a right light bar G1R is provided in a housing of the right microphone M1R. Further, a rear camera S6B and a rear microphone M1B are provided adjacent to the rear speaker SP1B, and a rear light bar G1B is provided in a housing of the rear microphone M1B.

[0151] With this configuration, the work machine **100** can turn on the light bar corresponding to the speaker that has been turned on (light emittable state), and turn off the light bar corresponding to the speaker that has been turned off (light unemittable state).

[0152] The external sound output device SP1 may be configured by one or a plurality of parametric speakers. Parametric speakers are ultrasonic speakers that can selectively transmit sound to people in a specific narrow range. Parametric speakers can transmit sound toward any location.

[0153] In the work machine **100** illustrated in FIG. **10**, the controller **30** may detect the worker WK around the work machine **100** based on an image captured by the imaging device S6 and specify a position of the worker WK. Further, when a plurality of workers WK are present around the work machine **100**, the controller **30** may discriminate between a conversation target person (a worker WK who is uttering voice) and a non-target person (a worker WK who is not uttering voice) based on the outputs of the four external sound collection devices M1. The controller **30** may distinguish between a conversation partner (the worker WK facing the work machine **100**) and a non-partner (the worker WK not facing the work machine **100**) based on the image captured by the imaging device S6. Then, the controller **30** may turn on the speaker and the light bar facing the worker WK. For example, when the worker WK (uttering voice) is present behind the work machine **100**, the controller **30** may turn on the rear speaker SP1B while keeping the front speaker SP1F, the left speaker SP1L, and the right speaker SP1R turned off. In this case, the controller **30** may turn on the rear light bar G1B while keeping the front light bar G1F, the left light bar G1L, and the right light bar G1R turned off. Such a function may be implemented in the work machine **100** illustrated in FIGS. **1** to **5**.

[0154] With this configuration, the controller **30** can output a sound in the direction in which the worker WK is present without outputting a sound in the direction in which the worker WK is absent. Therefore, the worker WK can easily recognize whether the worker WK is a conversation target person or a non-target person.

[0155] Next, a configuration example of the operation system SYS according to the embodiment of the present disclosure will be described with reference to FIG. **11**. FIG. **11** is a schematic view illustrating a configuration example of the operation system SYS. As illustrated in FIG. **11**, the operation system SYS includes the work machine **100**, a remote control room RC, and a

management center MC. Note that the detailed configuration of the work machine **100** is not illustrated in FIG. **11**. This is because the work machine **100** illustrated in FIG. **11** has the same configuration as the work machine **100** illustrated in FIG. **1** or FIG. **10**.

[0156] The work machine **100**, the remote control room RC, and the management center MC are connected to each other via a communication network NW so as to be able to transmit and receive data. The work machine **100**, the remote control room RC, and the management center MC may be connected to each other so as to be able to directly transmit and receive data to and from each other without passing through the communication network NW. In the illustrated example, the work machine **100** transmits information related to the work site to the remote control room RC. Thus, the remote operator RO in the remote control room RC can grasp the situation of the work site based on the information from the work machine **100**.

[0157] The work machine **100** is provided with a sensor capable of three dimensionally recognizing the position and shape of an object existing in the work site. For example, the work machine **100** is provided with a space recognition device. Therefore, the work machine **100** can transmit a result of a three dimensionally measured work site to the remote control room RC.

[0158] The space recognition device is a device for recognizing a space around the work machine **100**. In the illustrated example, the space recognition device is a LiDAR. The LiDAR measures, for example, a distance between the LiDAR and each of one million or more points within a monitoring range. The space recognition device may be any device that can measure the distance to an object. For example, the space recognition device may be a stereo camera, or may be a combination of the imaging device **S6** and a distance measuring device such as a millimeter wave radar.

[0159] The number of work machines **100** included in the operation system SYS may be one or more. In the case where a plurality of work machines **100** are included, the remote operator RO of a specific work machine **100** can acquire information related to the work site acquired by the specific work machine **100** and additionally acquire information related to the work site acquired by another one or more other work machines **100**.

[0160] The remote control room RC is provided with a communication device **T2**, a remote controller **40**, an operation device **42**, an operation sensor **43**, a display device **D1E**, an internal sound collection device **M2E**, and an internal sound output device **SP2E**. In addition, an operation seat **DS** on which a remote operator RO who remotely operates the work machine **100** sits is installed in the remote control room RC.

[0161] The communication device **T2** is configured to be able to communicate with the communication device **T1** attached to the work machine **100**.

[0162] The remote controller **40** is a computing device that executes various computations. In the present embodiment, the remote controller **40** is configured by a microcomputer including a CPU (Central Processing Unit) and a memory. The various functions of the remote controller **40** are implemented by the CPU executing programs stored in the memory.

[0163] The display device **D1E** is a device capable of displaying various kinds of information. The display device **D1E** displays an image based on information transmitted from the work machine **100** in order for the remote operator RO in the remote control room RC to visually recognize the surroundings of the work machine **100**. In the illustrated example, the display device **D1E** is a liquid-crystal display that displays an image captured by the imaging device **S6** mounted on the work machine **100**. The display device **D1E** may be a display or a projector that implements naked-eye stereoscopic vision, or may be a virtual reality (VR) goggle or the like.

[0164] The internal sound output device **SP2E** is a device capable of outputting various kinds of sound information. The internal sound output device **SP2E** outputs sound based on information transmitted from the work machine **100** so that the remote operator RO in the remote control room RC can hear sound emitted at the work site. The internal sound output device **SP2E** may be configured to output a sound captured by the external sound collection device **M1** attached to the

outside of the cabin **10**, or may be configured to output a sound captured by the internal sound collection device **M2** attached to the inside of the cabin **10**, for example. In this case, the internal sound collection device **M2** may be provided at a position corresponding to the position of the ears of the operator seated in the operator's seat **50** in the cabin **10**. The internal sound output device **SP2E** may be a stationary device such as a speaker, or may be a wearable device such as earphones or headphones. The speaker may be a monaural speaker, a stereo speaker, or a surround speaker. The speaker may be an omnidirectional speaker or a directional speaker. The wearable device may have a noise canceling function, a spatial audio function (stereophonic function), or a bone conduction function.

[0165] The operation device **42** is provided with the operation sensor **43** for detecting an operation content of the operation device **42**. The operation sensor **43** is, for example, an inclination sensor that detects an inclination angle of the operation lever, an angle sensor that detects a slewing angle of the operation lever around a slewing axis, or the like. The operation sensor **43** may be configured by another sensor such as a pressure sensor, a current sensor, a voltage sensor, or a distance sensor. The operation sensor **43** outputs information related to the detected operation content of the operation device **42** to the remote controller **40**. The remote controller **40** generates an operation signal based on the received information, and transmits the generated operation signal to the work machine **100**. The operation sensor **43** may be configured to generate the operation signal. In this case, the operation sensor **43** may output the operation signal to the communication device **T2** without passing through the remote controller **40**. With such a configuration, the remote operator **RO** can remotely operate the work machine **100** from the remote control room **RC**.

[0166] The management center **MC** is a facility in which various devices for managing the work machine **100** at the work site or the remote operation of the work machine **100** by the remote operator **RO** in the remote control room **RC** are provided. In the illustrated example, the management center **MC** is installed at a position away from each of the work site of the work machine **100** and the remote control room **RC**. The management center **MC** is provided with a management device **200**, an internal sound collection device **M2C**, and an internal sound output device **SP2C**.

[0167] The management device **200** is an example of a control device, and is, for example, a server computer (so-called cloud server) or an edge server. The management device **200** is typically a fixed terminal device, but may be a mobile terminal device (e.g., a laptop computer, a tablet, or a smartphone).

[0168] With this configuration, the manager in the management center **MC** can hear the sound emitted at the work site by using the sound collection device (the external sound collection device **M1** or the internal sound collection device **M2**) attached to the work machine **100** and the internal sound output device **SP2C**, for example. The manager in the management center **MC** can hear the sound generated in the remote control room **RC** by using the internal sound collection device **M2E** and the internal sound output device **SP2C** provided in the remote control room **RC**, for example. Further, the manager at the management center **MC** can transmit his/her voice to the worker **WK** around the work machine **100** by using, for example, the internal sound collection device **M2C** and the external sound output device **SP1** attached to the work machine **100**. The manager at the management center **MC** can also use the internal sound collection device **M2C** and the internal sound output device **SP2** attached to the work machine **100** to transmit his/her voice to the operator **OP** of the work machine **100**. The manager in the management center **MC** can transmit his/her voice to the remote operator **RO** in the remote control room **RC** by using the internal sound collection device **M2C** and the internal sound output device **SP2E** provided in the remote control room **RC**.

[0169] As described above, the work machine **100** according to the embodiment of the present disclosure includes the cabin **10**, the external sound collection device **M1** disposed outside the cabin **10**, the internal sound output device **SP2** disposed inside the cabin **10**, and the information

transmission device **G1** that enables the worker **WK** to recognize whether a voice of the worker **WK** around the work machine **100** has reached the operator **OP** of the work machine **100**.

[0170] This configuration provides an effect that the information transmitted by the information transmission device **G1** enables the worker **WK** to recognize that the voice of the worker **WK** around the work machine **100** has reached the operator **OP** of the work machine **100**.

[0171] For example, when the information transmission device **G1** is not mounted on the work machine **100**, the worker **WK** cannot see the reaction of the operator **OP** who is hidden behind the upper slewing body **3** and cannot confirm whether or not his/her voice has reached the operator **OP** in a situation where the worker **WK** speaks to the operator **OP** from the behind of the work machine **100**. In a situation where the worker **WK** cannot confirm whether or not his/her voice has reached the operator **OP**, the worker **WK** may erroneously considers that his/her voice has reached the operator **OP**. In this case, although the voice of the worker **WK** has not actually reached the operator **OP**, the worker **WK** may misunderstand that the operator **OP** is aware of the presence of the worker **WK** and may carelessly approach the work machine **100**. In addition, when the operator **OP** is spoken by the worker **WK**, the operator **OP** notifies the worker **WK** of the fact that the operator **OP** has noticed the presence of the worker **WK** by a gesture, and thus the operator **OP** may interrupt the work. Therefore, a configuration in which the information transmission device **G1** is not mounted on the work machine **100** may reduce the work performance of the work machine **100**.

[0172] When the information transmission device **G1** is mounted on the work machine **100**, the worker **WK** behind the work machine **100** can check whether or not his/her own voice has reached the operator **OP** by looking at a light emission state of the information transmission device **G1** (rear light bar **G1B**). Therefore, this configuration can prevent the occurrence of a situation in which the worker **WK** erroneously considers that his/her voice has reached the operator **OP** even though the voice of the worker **WK** has not actually reached the operator **OP**, and can prevent the worker **WK** from inadvertently approaching the work machine **100**. Further, this configuration also makes it possible to prevent the operator **OP** from being forced to take his/her hand off the operation device **26** and interrupt the operation to perform a gesture.

[0173] The information transmission device **G1** may be configured to be able to transmit an activation state of the information transmission device **G1** to the worker **WK**. The activation state may include, for example, an ON state, an OFF state, a state in which the voice of the operator **OP** is being collected, or a state in which the voice of the operator **OP** is collectable. The state in which the voice of the operator **OP** is collectable is, for example, a state in which the operator **OP** presses the speech button **KS**.

[0174] With this configuration, the worker **WK** can accurately check various situations such as whether or not his/her voice is transmitted to the operator **OP** when the worker **WK** speaks to the operator **OP**, whether or not the operator **OP** is attempting to speak to the worker **WK**, and whether or not the operator **OP** is uttering a voice. Therefore, this configuration provides an advantageous effect of preventing a hindrance in conversations (conversations that are more difficult to communicate each other than face-to-face conversations) between the worker **WK** outside the cabin **10** and the operator **OP** inside the cabin **10**.

[0175] The work machine **100** may include the internal sound collection device **M2** disposed inside the cabin **10** and an external sound output device **SP1** disposed outside the cabin **10**.

[0176] This configuration provides an effect that the voice of the operator **OP** in the cabin **10** can be output toward the worker **WK** outside the cabin **10** at an appropriate volume. Therefore, this configuration eliminates the need for the operator **OP** to speak to the worker **WK** with a loud voice, and can reduce the burden on the operator **OP** when having a conversation with the worker **WK**.

[0177] The work machine **100** may include a speech button **KS** disposed in the cabin **10**. The work machine **100** may be configured such that, when the speech button **KS** is pressed, the internal sound collection device **M2** collects sound in the cabin **10**, and the external sound output device **SP1** outputs the sound collected by the internal sound collection device **M2** toward the

surroundings of the work machine **100**.

[0178] This configuration can prevent the sound generated in the cabin **10** from being continuously output to the outside of the cabin **10** while allowing the operator OP to output his/her voice to the worker WK outside the cabin **10** as necessary, and thus can enhance the privacy of the operator OP.

[0179] The information transmission device **G1** is preferably provided adjacent to the external sound collection device **M1**. This configuration provides an effect of enabling the worker WK to more easily recognize that the voice of the worker WK has reached the operator OP of the work machine **100**. This is because the worker WK typically directs his/her eyes on the external sound collection device **M1** when speaking toward the external sound collection device **M1**, and at this time, if the information transmission device **G1** is adjacent to the external sound collection device **M1**, the information transmission device **G1** naturally comes into the eyes.

[0180] The operation system **SYS** according to the embodiment of the present disclosure includes the external sound collection device **M1** attached to the work machine **100**, an internal sound output device **SP2** (internal sound output device **SP2E**) that outputs sound to the operator OP (remote operator RO) of the work machine **100**, and the information transmission device **G1** that enables the worker WK to recognize whether the voice of the worker WK around the work machine **100** has reached the operator OP (remote operator RO).

[0181] This configuration provides an effect that the information transmitted by the information transmission device **G1** enables the worker WK to recognize that the voice of the worker WK around the work machine **100** has reached the operator OP. This configuration also provides an effect that even when the work machine **100** is remotely controlled, the information transmitted by the information transmission device **G1** enables the worker WK to recognize that the voice of the worker WK around the work machine **100** has reached the remote operator RO.

[0182] The work machine **100** according to the embodiment of the present disclosure includes the internal sound collection device **M2** that collects a voice of the operator OP of the work machine **100**, the external sound output device **SP1** attached to the work machine **100**, and the controller **30** as a control device that detects that the operator OP of the work machine **100** is attempting to have a conversation with a person (worker WK) around the work machine **100** using the internal sound collection device **M2** and the external sound output device **SP1** and notifies the surroundings of the work machine **100** of the detection fact.

[0183] This configuration provides an effect that the worker WK can recognize that the operator OP of the work machine **100** is attempting to speak to the worker WK. In this configuration, even in a situation where a plurality of work machines are activating at the work site, the worker WK can recognize which operator of the work machine is attempting to speak to the worker WK. In this configuration, the worker WK can recognize that the operator OP is attempting to speak to the worker WK before the operator OP utters. Therefore, this configuration can prevent the worker WK from speaking at the same timing as the timing at which the operator OP speaks. Therefore, the worker WK can speak at a timing different from the timing at which the operator OP speaks.

[0184] The controller **30** may be configured to be able to audibly, tactually, or visually notify a person around the work machine **100** that the operator OP of the work machine **100** is attempting to have a conversation with a person around the work machine **100** (the worker WK) before the operator OP of the work machine **100** utters or during the utterance of the operator OP of the work machine **100**.

[0185] This configuration provides an effect that the operator OP can notify the worker WK at an appropriate timing that the operator OP is attempting to speak to the worker WK.

[0186] The work machine **100** may include a speech button **KS** disposed in the cabin **10**. In this case, when the speech button **KS** is pressed, the internal sound collection device **M2** may collect sound in the cabin **10**, and the external sound output device **SP1** may output the sound collected by the internal sound collection device **M2** toward the surroundings of the work machine **100**. Then, when the speech button **KS** is pressed, the controller **30** may notify those on the outside of the

cabin **10** that the operator OP of the work machine **100** is attempting to have a conversation with a person (worker WK) outside the cabin **10**.

[0187] This configuration provides an effect that the operator OP can make the worker WK recognize that the operator OP is about to speak to the worker WK.

[0188] The controller **30** may be configured not to perform notification until a predetermined time elapses after the previous notification is performed. For example, the controller **30** may be configured not to perform the auditory notification until a predetermined time elapses after the previous auditory notification is performed.

[0189] This configuration provides an effect of preventing the notification from being frequently performed and causing the worker WK to feel bothersome. However, the controller **30** may be configured to output at least one of the visual notification and the tactile notification even when the predetermined time has not elapsed after the previous notification. This is because the visual notification and the tactile notification are less likely to cause the worker WK to feel bothersome than the auditory notification even if the visual notification and the tactile notification are frequently output.

[0190] Further, the notification to the surroundings of the work machine **100** may be implemented by outputting a predetermined sound. In this case, the predetermined sound is preferably a sound distinguished from the horn sound.

[0191] This configuration provides an effect that the operator OP can notify the worker WK that the operator OP is attempting to have a conversation with the worker WK without outputting a horn sound having a high effect of attracting attention, that is, without excessively attracting attention of a person around the work machine **100**.

[0192] The work machine **100** may include the internal sound output device SP2 disposed inside the cabin **10** and the external sound collection device M1 disposed outside the cabin **10**.

[0193] This configuration provides an effect that the voice uttered by the worker WK can be more reliably transmitted to the operator OP after the operator OP notifies the worker WK that the operator OP is attempting to have a conversation with the worker WK. This is because the voice of the worker WK collected by the external sound collection device M1 is reproduced by the internal sound output device SP2.

[0194] Further, the controller **30** may be configured to disable the internal sound collection device M2 from collecting sound related to the auditory notification when the auditory notification is being performed. For example, the controller **30** may turn off the internal sound collection device M2 (a state in which sound collection is not possible) when an auditory notification is being made.

[0195] This configuration provides an effect of preventing the occurrence of howling. In particular, this configuration provides an effect that it is possible to prevent the sound (auditory notification) output by the external sound output device SP1 from being collected by the internal sound collection device M2 and further the sound (auditory notification) collected by the internal sound collection device M2 from being output by the external sound output device SP1.

[0196] The work machine **100** may include a human detection device that detects a human around the work machine **100**. The human detection device may be configured by the imaging device S6, LiDAR, or the like. In this case, the controller **30** may be configured to, when detecting that the operator OP of the work machine **100** is attempting to have a conversation with a person (worker WK) around the work machine **100**, identify the position of the person around the work machine **100** based on the output of the human detection device and notify the operator OP of the fact (i.e., the fact that the operator OP of the work machine **100** is attempting to have a conversation with a person (worker WK) around the work machine **100**) in the direction in which the person (worker WK) is present.

[0197] This configuration provides an effect that the specific worker WK can be notified that the operator OP is attempting to speak. That is, this configuration provides an effect that the specific worker WK can be notified that the operator OP is attempting to speak without attracting attention

of a person other than the specific worker WK.

[0198] The controller **30** may be configured to output a sound toward the direction in which the person (worker WK) is present, or to cause the light emitter to emit light toward the direction in which the person (worker WK) is present.

[0199] This configuration provides an effect that the specific worker WK can be more reliably notified that the operator OP is attempting to speak.

[0200] The operation system SYS according to the embodiment of the present disclosure includes the internal sound collection device M2 (internal sound collection device M2E) that collects voices of the operator OP (remote operator RO) of the work machine **100**, the external sound output device SP1 attached to the work machine **100**, and the controller **30** as a control device that detects that the operator OP (remote operator RO) of the work machine **100** is attempting to have a conversation with a person (worker WK) around the work machine **100** using the internal sound collection device M2 (internal sound collection device M2E) and the external sound output device SP1 and notifies the surroundings of the work machine **100** of the detection.

[0201] This configuration provides an effect of enabling the worker WK to recognize that the operator OP of the work machine **100** is attempting to speak to the worker WK. This configuration also provides an effect of enabling the worker WK to recognize that the remote operator RO of the work machine **100** is attempting to speak to the worker WK.

[0202] The preferred embodiments of the present disclosure have been described above. However, the invention according to the present disclosure is not limited to the above-described embodiments. Various modifications, substitutions, and the like can be applied to the above-described embodiments without departing from the scope of the invention according to the present disclosure. Further, the features described with reference to the above-described embodiments may be appropriately combined as long as there is no technical contradiction.

Claims

1. A work machine comprising: a lower traveling body; an upper slewing body slewably mounted on the lower traveling body; an external microphone configured to collect sound around a work machine; an internal speaker configured to output sound to an operator of the work machine; an information transmission device configured to enable a worker around the work machine to recognize whether or not a voice of the worker has reached the operator of the work machine; and a processor, and a memory storing instructions that cause the processor to execute a process, wherein the process includes collecting, by the external microphone, sound around the work machine; outputting, by the internal speaker, the sound to the operator of the work machine; and enabling, by the information transmission device, the worker around the work machine to recognize whether or not a voice of the worker has reached the operator of the work machine.
2. The work machine according to claim 1, wherein the enabling includes transmitting an activation state of the information transmission device to the worker, and wherein the activation state includes an ON state, an OFF state, a state in which a voice of the operator of the work machine is collected, or a state in which the voice of the operator of the work machine is collectable.
3. The work machine according to claim 1, further comprising: an internal microphone configured to collect a voice of the operator of the work machine; and an external speaker configured to output sound toward surroundings of the work machine, wherein the process further includes collecting, by the internal microphone, a voice of the operator of the work machine; and outputting, by the external speaker, the sound toward the surroundings of the work machine.
4. The work machine according to claim 3, further comprising: a speech button, wherein the internal microphone collects sound when the speech button is pressed, and the external speaker outputs the sound collected by the internal microphone toward the surroundings of the work machine.

5. The work machine according to claim 1, wherein the information transmission device is provided adjacent to the external microphone.

6. An operation system for a work machine, the operation system comprising: an external microphone configured to collect sound around a work machine; an internal speaker configured to output sound to an operator of the work machine; an information transmission device configured to enable a worker around the work machine to recognize whether or not a voice of the worker has reached the operator of the work machine; and a processor, and a memory storing instructions that cause the processor to execute a process, wherein the process includes collecting, by the external microphone, sound around the work machine; outputting, by the internal speaker, the sound to the operator of the work machine; and enabling, by the information transmission device, the worker around the work machine to recognize whether or not a voice of the worker has reached the operator of the work machine.
